ROUH
Supporting Smarter IVF Decisions

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Smart AI-Based System for Non-Invasive Embryo Selection in IVF Procedures

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Abstract

ROUH, an AI-powered decision support system for embryo viability assessment in IVF. By combining embryo image analysis and morphokinetic developmental patterns through a late fusion strategy, the system provides transparent and non-invasive viability predictions.

Objectives

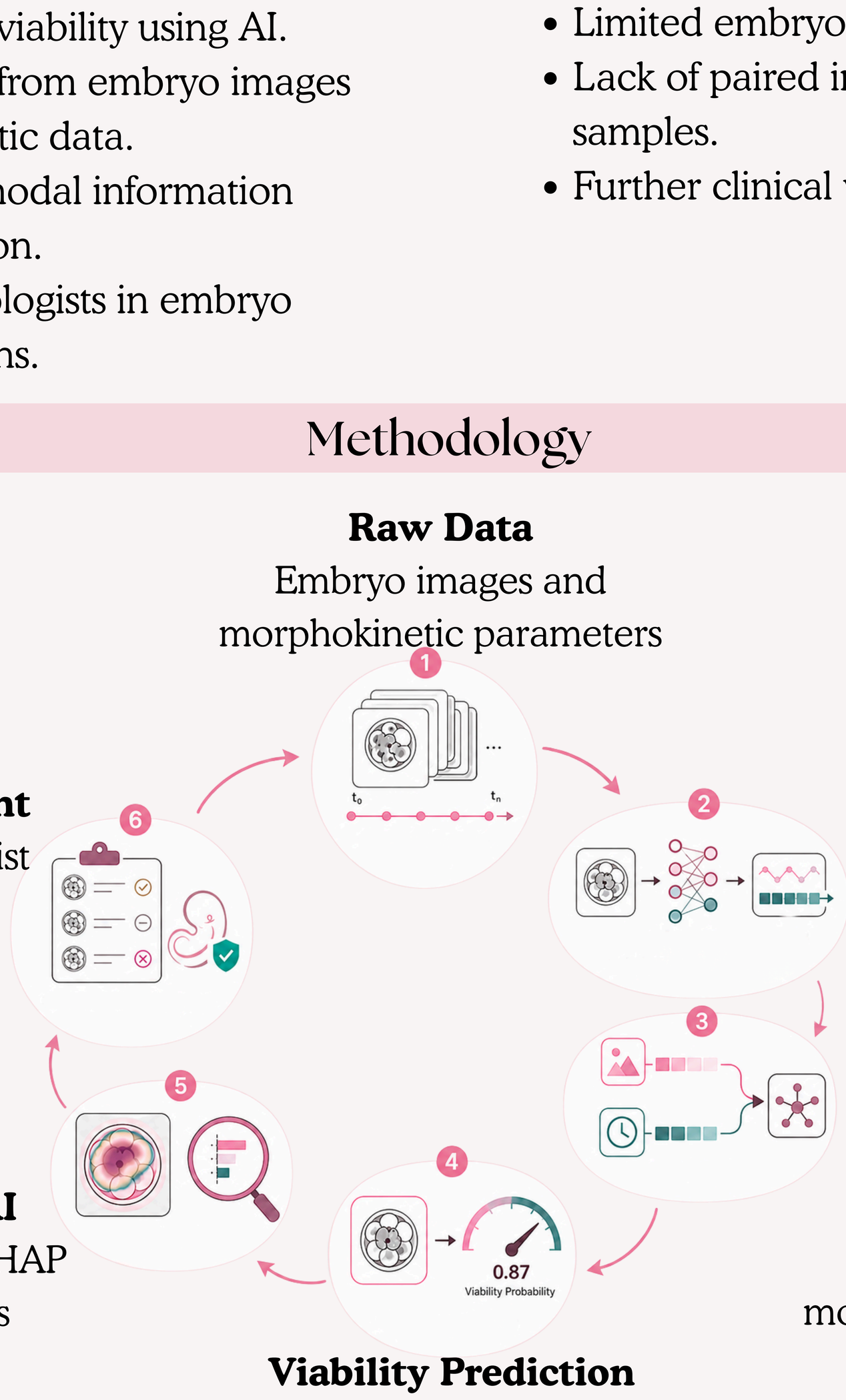
- Predict embryo viability using AI.
- Extract features from embryo images and morphokinetic data.
- Integrate multimodal information through late fusion.
- Support embryologists in embryo selection decisions.

Limitations

- Limited embryo dataset.
- Lack of paired image–timeline samples.
- Further clinical validation required.

Methodology

Raw Data
Embryo images and morphokinetic parameters



Feature Extraction
EfficientNetB0 and LSTM

Late Fusion
Combine visual and morphokinetic probability

Viability Prediction
Late Fusion module generates probability

Explainable AI
Grad-CAM and SHAP provide insights

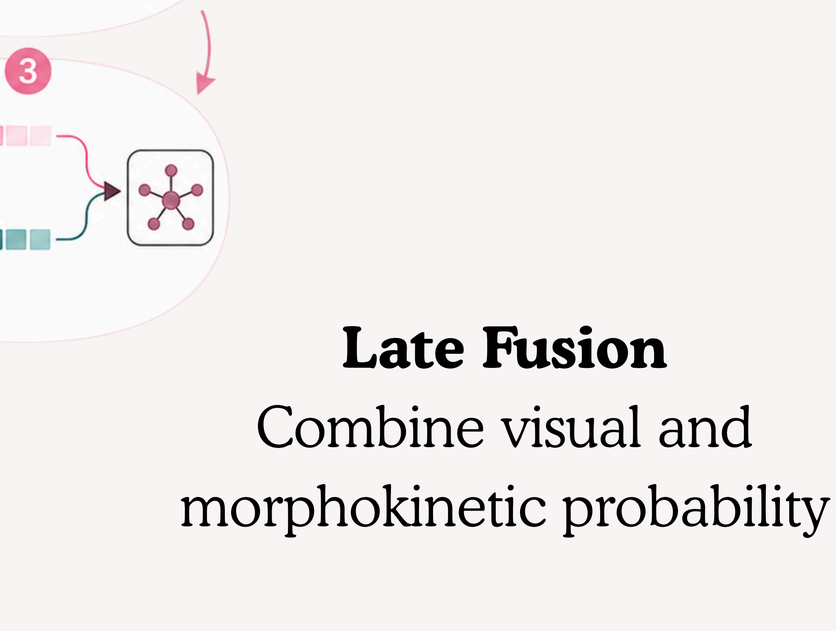
Embryo Assessment
Supports embryologist embryo selection

Conclusion

Image Model
Generates image-based probability

Morphokinetic Model
Generates viability score

Late Fusion
Combines both model outputs



Future Works

Future work will focus on larger datasets, clinical data integration, improved fusion methods, and real-world validation.